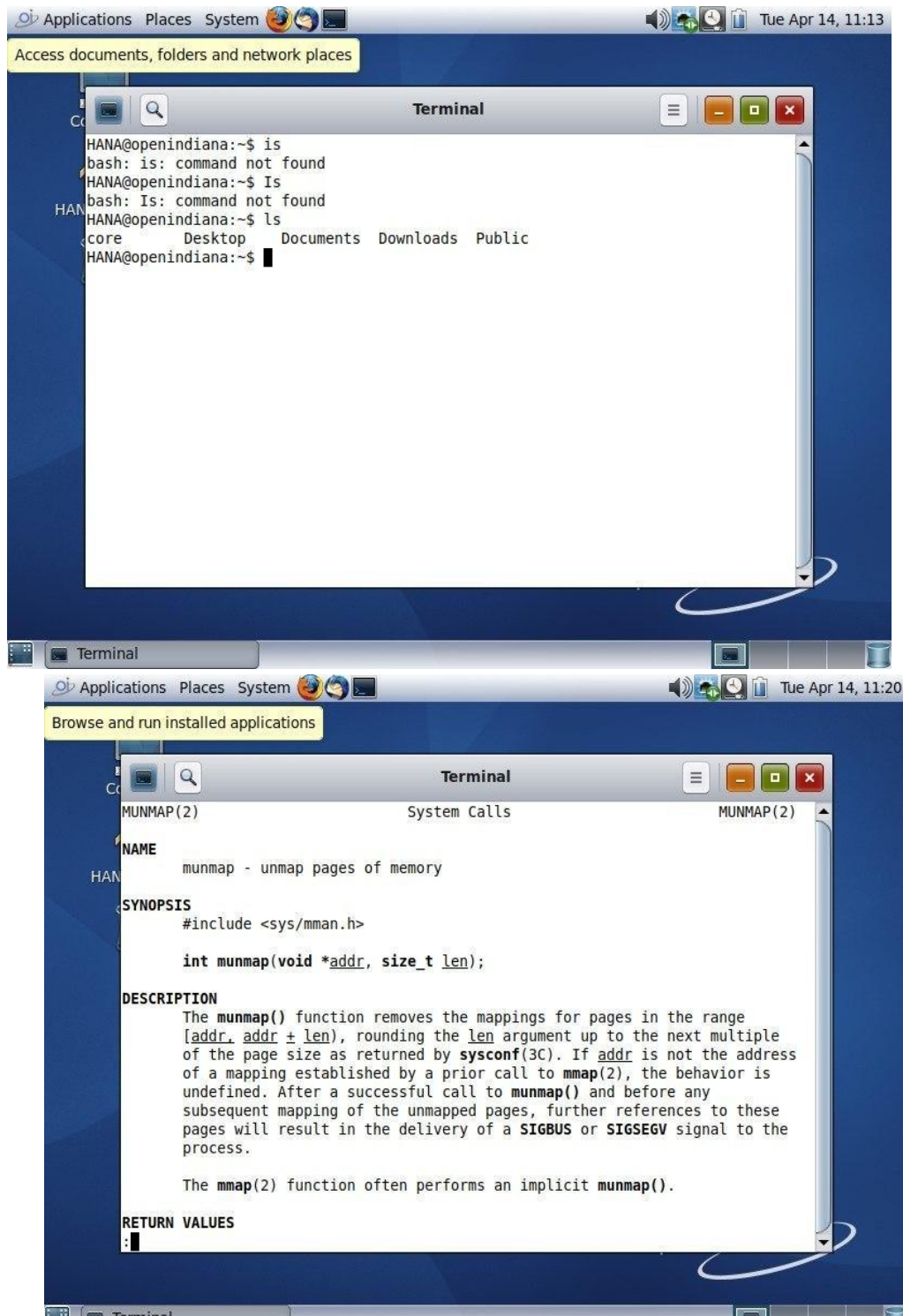


11. System call implementation

System calls are the interface between user programs and the operating system. In OpenIndiana, system calls allow applications to request services from the kernel, such as file handling, process control, and memory management.

In OpenIndiana, system calls are implemented in a layered way. When a user program wants to perform an operation (for example, reading a file), it does not directly access the hardware. Instead, it sends a request to the operating system through a system call.



```
HANA@openindiana:~$ is
bash: is: command not found
HANA@openindiana:~$ Is
bash: Is: command not found
HANA@openindiana:~$ ls
core  Desktop  Documents  Downloads  Public
HANA@openindiana:~$
```

```
MUNMAP(2)                                System Calls                                MUNMAP(2)

NAME
    munmap - unmap pages of memory

SYNOPSIS
    #include <sys/mman.h>

    int munmap(void *addr, size_t len);

DESCRIPTION
    The munmap() function removes the mappings for pages in the range
    [addr, addr + len), rounding the len argument up to the next multiple
    of the page size as returned by sysconf(3C). If addr is not the address
    of a mapping established by a prior call to mmap(2), the behavior is
    undefined. After a successful call to munmap() and before any
    subsequent mapping of the unmapped pages, further references to these
    pages will result in the delivery of a SIGBUS or SIGSEGV signal to the
    process.

    The mmap(2) function often performs an implicit munmap().

RETURN VALUES
    :
```

Change desktop appearance and behavior, get help, or log out

```
mouse, 1instance g0
p1t_beep, Instance â6
d1spTay, 1instance #D
pc18086, be, Instance #B
pci BDee, cafe (dr1ver not attached)
pc18686, 8, 1instance GB
pcilB6b, 3f, instance #B
input, instance #0
pciB0B6, 7113 (driver not attached)
pci808b.265c, instance #0
pciBoB6, 2B29, instance #o
disk, instance #0
cdrom, 1instance ll
fu, 1instance #6
cpu (d r1ver not attached)
cpu (d r1ver not attached)
sb, Ins tance 61
useJresources (driver not attached)
iscsi, instance #8
agpgart, instance80 (driver not attached)
options. instance #0
pseudo, instance 88
xsvc, instance #0
NANgopenndana:-$
```

Browse and run installed applications

```
mouse, 1instance g0
p1t beep, Instance #6
d1splay, 1instance #o
pciB0Bb, 1e, instance #0
pc18Dee, cafe (dr1ver not attached)
pciB6B6, 8, 1instance GB
pet LB6b, 3f, Instance GB
input, 1instance gD
pc18086, 7113 (dr1ver not attached)
pciB0Bb, 265c, instance #8
pc18D86, 2829, Instance #B
d1sk, Instance #0
cdrom, 1instance gl
fu, Instance #8
cpu (d r1ver not attached)
cpu (driver not attached)
sb, Instance #1
used-resources (dr1ver not attached)
1scs1, 1instance GB
aqpgart, 1instance TO (d r1ver not attached)
opt1ons, 1instance gEi
pseudo, Instance g8
xsvc, instance #6
HA5Agopen1ndlana:-$ vmstat
```

```

pci80ee,cafe (driver not attached)
pci8086,0, instance #0
pci106b,3f, instance #0
    input, instance #0
pci8086,7113 (driver not attached)
pci8086,265c, instance #0
pci8086,2829, instance #0
    disk, instance #0
    cdrom, instance #1
fw, instance #0
    cpu (driver not attached)
    cpu (driver not attached)
    sb, instance #1
used-resources (driver not attached)
iscsi, instance #0
agpgart, instance #0 (driver not attached)
options, instance #0
pseudo, instance #0
xsvc, instance #0
HANA@openindiana:~$ vmstat
kthr      memory          page            disk           faults        cpu
 r  b w   swap free  re  mf pi po fr de sr ro s0 s1 --  in  sy  cs us sy id
0  0  0 246896 133360 135 370 0  0  9  0 20742 14 14 -1 0 1019 2096 834 1 26 73
HANA@openindiana:~$

```

```

HANA@openindiana:~$ vmstat
kthr      memory          page            disk           faults        cpu
 r  b w   swap free  re  mf pi po fr de sr ro s0 s1 --  in  sy  cs us sy id
0  0  0 246896 133360 135 370 0  0  9  0 20742 14 14 -1 0 1019 2096 834 1 26 73
HANA@openindiana:~$ munmap/usr/include/sys/mman.h
bash: munmap/usr/include/sys/mman.h: No such file or directory
HANA@openindiana:~$ grep ""
>
> /usr/include
>
> clear
> clere
> grep "munmap" /usr/include/sys/mman.h
>
> ^C
HANA@openindiana:~$ grep "munmap" /usr/include/sys/mman.h
grep: can't open "/usr/include/sys/mman.h"
HANA@openindiana:~$ ls /usr/include/sys
dcam          fasttrap.h    lockstat.h    sdt.h
dtrace_impl.h fct_defines.h lpif.h         stmf_defines.h
dtrace.h      fct.h         mdb_modapi.h  stmf_ioctl.h
fasttrap_impl.h fctio.h       nvme           stmf.h
fasttrap_isa.h ieeeep.h      portif.h
HANA@openindiana:~$

```

Applications Places System        Tue Apr 14, 11:39

Change desktop appearance and behavior, get help, or log out

```

r b w  swap free re mf pi po fr de sr ro s0 s1 -- in sy cs us sy id
0 0 0 246896 133360 135 370 0 0 9 0 20742 14 14 -1 0 1019 2096 834 1 26 73
HANA@openindiana:~$ munmap/usr/include/sys/mman.h
bash: munmap/usr/include/sys/mman.h: No such file or directory
HANA@openindiana:~$ grep ""
>
> /usr/include
>
> clear
> clere
> grep "munmap" /usr/include/sys/mman.h
>
> ^C
HANA@openindiana:~$ grep "munmap" /usr/include/sys/mman.h
grep: can't open "/usr/include/sys/mman.h"
HANA@openindiana:~$ ls /usr/include/sys
dcam          fasttrap.h    lockstat.h    sdt.h
dtrace_impl.h fct_defines.h lpif.h         stmf_defines.h
dtrace.h      fct.h         mdb_modapi.h  stmf_ioctl.h
fasttrap_impl.h fctio.h       nvme           stmf.h
fasttrap_isa.h ieeeep.h      portif.h
HANA@openindiana:~$ pagesize
4096
HANA@openindiana:~$
```



```

00007FFFAF2E0000    1524K r-x-- /lib/amd64/libc.so.1
00007FFFAF46D000     48K rw--- /lib/amd64/libc.so.1
00007FFFAF479000     16K rw--- /lib/amd64/libc.so.1
00007FFFAF48F000      4K rwxS- [ anon ]
00007FFFAF4A0000     64K rwx-- [ anon ]
00007FFFAF4C0000    128K rwx-- [ anon ]
00007FFFAF4EB000     80K r---- /usr/lib/locale/en_US.UTF-8/LC_COLLATE/LCL_D
ATA
00007FFFAF500000     24K rwx-- [ anon ]
00007FFFAF510000      4K rwx-- [ anon ]
00007FFFAF520000      4K rwx-- [ anon ]
00007FFFAF530000      4K rwx-- [ anon ]
00007FFFAF540000      4K rwx-- [ anon ]
00007FFFAF550000      4K rw--- [ anon ]
00007FFFAF560000      4K rw--- [ anon ]
00007FFFAF570000      4K rwx-- [ anon ]
00007FFFAF580000      4K rwx-- [ anon ]
00007FFFAF590000      8K r--S- [ comm ]
00007FFFAF599000    324K r-x-- /lib/amd64/ld.so.1
00007FFFAF5FA000     12K rwx-- /lib/amd64/ld.so.1
00007FFFAF5FD000      8K rwx-- /lib/amd64/ld.so.1
00007FFFBFFFA000     24K rw--- [ stack ]
total    4588K
HANA@openindiana:~$

```

12. Over all personal experiance

This project was about installing and using OpenIndiana in a virtual machine using Oracle VM VirtualBox. It helped to understand how operating systems are installed, configured, and managed in a safe virtual environment.

During the work, I learned basic system setup, virtualization concepts, and how UNIX-based systems operate. I also faced and solved common installation issues, which improved my practical skills.

OpenIndiana uses the ZFS filesystem and follows UNIX standards, making it stable and reliable for learning and system practice.

13. References

- OpenIndiana Official Documentation – <https://www.openindiana.org8>
- Oracle VM VirtualBox User Guide – <https://www.virtualbox.org/manual8>

- [UNIX System and POSIX Standards Overview – IEEE Standards Association](#)